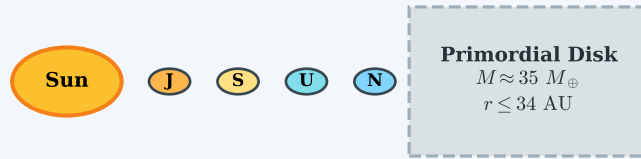
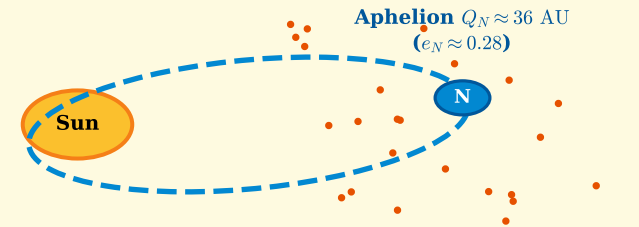


1. Primordial Architecture & Disk Truncation



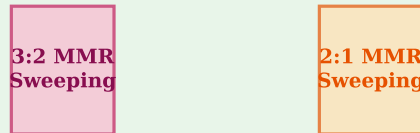
Outer Solar System formed in compact resonant state.
 Massive planetesimal disk truncated at $r_{\text{edge}} \approx 30 - 34 \text{ AU}$.
 Current Kuiper belt region ($r > 40 \text{ AU}$) was initially empty!

2. Dynamical Instability & Neptune Scattering



Close encounter with Uranus scatters Neptune outward to $\sim 28 \text{ AU}$
 with large transient eccentricity ($e_N \approx 0.28$).
 Exterior resonances (3:2, 2:1) sweep through the primordial disk.

3. Resonant Sweeping & Chaotic Transport



Sweeping resonances drag and excite planetesimals outward.
 Mean-motion resonance overlap causes chaotic diffusion into 42 – 48 AU.
 Eccentricity is pumped to $e \sim 0.1 - 0.3$.

4. Decoupling & Cold/Hot Population Origin



Dynamical friction damps Neptune eccentricity ($\tau_d \approx 3 \text{ Myr}$).
 Neptune radial reach shrinks $\rightarrow$ planetesimals with $q > 36 \text{ AU}$ decouple!
 Populations frozen permanently: Cold ($i < 4^\circ$) & Hot ($i \sim 14^\circ$).